

From Spectra to Topology

The Hodge–Dijkgraaf–Witten bridge: harmonic qubits, holonomy,
the Choi bend, and finding geometry by minimizing eigenvector residuals

tessera — cobordism realizability programme

Abstract

This primer assembles the theory behind the realizability bridge. We are after a single statement of the *state–operation–cobordism correspondence*: a quantum operation U is realizable as a bulk cobordism W_{AB} whose value reproduces the transition amplitude $\langle\psi_A|U|\psi_B\rangle$. There are two roads to that value. The *spectral* road builds the geometry: states become harmonic forms of a combinatorial Hodge Laplacian, and we *synthesize* a bulk whose boundary Laplacian carries the (Choi-bent) operator as an eigenvector. The *topological* road is the Dijkgraaf–Witten state sum: a metric-free invariant computed from the cobordism’s topology and a group cocycle. The bridge asks whether the two roads give the same number. We develop, in order: the combinatorial Hodge Laplacian and its harmonic spaces; the *Hodge qubit*; flat $\mathbb{Z}_2$ holonomy and the Dijkgraaf–Witten partition function; the crucial distinction between the b_1 -dimensional harmonic space and the 2^{b_1} -dimensional holonomy space, and the preparation map between them; the Choi–Jamiołkowski “bend” that turns an operation into a state; and a whitepaper-style algorithm that recovers *topology* by driving an eigenvector residual to zero.

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1 The realization map $\text{geo}(\cdot)$

Informally, geo is the *realization* (synthesis) map that turns quantum data into geometry — the inverse of the usual “read a spectrum off a given complex” pipeline. It acts on two kinds of input.

Definition 1 (The realization map). *On states.* For a normalized quantum state ψ , $\text{geo}(\psi)$ is a closed oriented *weighted* simplicial surface Σ that carries ψ as a harmonic form: $\psi \in \ker L_1(\text{geo}(\psi))$. The amplitudes and relative phase of ψ are encoded as the edge weights and holonomies of that harmonic 1-form (§4). A single qubit realizes as the torus ($g = 1$, $b_1 = 2$); a g -qubit state as a genus- g surface.

On operations. For an operation $U : \mathcal{H}_B \rightarrow \mathcal{H}_A$, $\text{geo}(U)$ is a cobordism (a manifold-with-boundary) whose interior weighted triangulation carries the Choi-bent operator $\text{vec}(U)$ (§7) as a boundary harmonic; this is the bulk W_{AB} of the Overview.

Four properties pin down how geo behaves; they are what the later sections silently assume.

- (G1) **Boundary-compatibility.** Restriction of an operation to its boundary states commutes with geo : $\partial \text{geo}(U) = \overline{\text{geo}(\psi_B)} \sqcup \text{geo}(\psi_A)$, the outgoing geometry of ψ_A together with the orientation-reversed incoming geometry of ψ_B . (This is hypothesis (H2) of the Overview.) So the state-level and operation-level halves of Definition 1 agree on shared boundaries.
- (G2) **Metric-laden.** geo outputs a *weighted* complex — edge weights and $U(1)$ phases, i.e. a metric and a connection, not just a triangulation. The spectral value Z_{spec} depends on these weights, in contrast to the Dijkgraaf–Witten road (§5), which discards them.
- (G3) **Defined implicitly, by spectral realization.** There is no closed formula for geo . A complex W is $\text{geo}(U)$ exactly when the eigenvector residual $r(W)$ of §8 vanishes — when the bent target is an eigenvector of the boundary Hodge Laplacian. The inverse problem of §8 *is* the constructive content of geo ; the realizability oracle is the search that evaluates it.
- (G4) **Partial and non-unique.** $\text{geo}(U)$ need not exist: it is defined only on the *realizable* operations. For an obstructed U the residual floors above zero and $\text{geo}(U)$ is undefined — the floor is the non-existence certificate (§8). When it does exist it is not unique: gauge/metric freedom and boundary-fixed Pachner moves yield many witnesses with $r(W) = 0$, and $\text{geo}(U)$ denotes any one of them.

With geo in hand the three hypotheses of the next section read directly: (H1) $\text{geo}(U)$ is a cobordism, (H2) its boundary is geo of the two states (G1), and (H3) its invariant is the amplitude. In the code-base geo on operations is realized by `EigenstateSynthesis / RealizabilityOracle.decide()` (§8) with the bend supplied by `ChoiJamiolkowski` (§7); the witness is returned as a `Spacetime` and `getBoundary()` recovers $\partial \text{geo}(U)$.

2 Overview: two roads to $Z(W_{AB})$

Fix two boundary states ψ_A, ψ_B and an operation $U : \mathcal{H}_B \rightarrow \mathcal{H}_A$. The correspondence claims three things about a bulk W_{AB} built “from” U :

- (H1) $W_{AB} = \text{geo}(U)$ is a cobordism (a manifold-with-boundary);
- (H2) $\partial W_{AB} = \overline{\text{geo}(\psi_B)} \sqcup \text{geo}(\psi_A)$ — its boundary is the (outgoing) geometry of ψ_A and the orientation-reversed (incoming) geometry of ψ_B ;
- (H3) $Z(W_{AB}) = \langle \psi_A | U | \psi_B \rangle$ — the cobordism’s invariant equals the transition amplitude.

The subtlety lives in (H3), because “ Z ” has two meanings:

- **Spectral / geometric Z .** Realize W_{AB} by *synthesizing* it: pin its boundary, fill its interior (edge weights and topology) until the boundary Hodge–Laplacian eigenvector matches the bent operator. The value is read off the geometry; it depends on the metric (edge weights).
- **Topological Z (Dijkgraaf–Witten).** Compute the $\mathbb{Z}_2$ state sum from the triangulation and a cocycle alone — no edge weights. The value is a topological invariant.

The bridge is the claim that these coincide on W_{AB} . The rest of this primer builds the machinery on both sides and the map between their boundary data.

3 The combinatorial Hodge Laplacian

Let K be a (finite, oriented) simplicial complex with weighted edges. Write C^k for the space of $\mathbb{C}$ -valued k -cochains (functions on oriented k -simplices, antisymmetric under reorientation). The *coboundary* $d_k : C^k \rightarrow C^{k+1}$ is the signed incidence operator,

$$(d_k \alpha)(\sigma) = \sum_{\tau \subset \sigma, \dim \tau = k} [\sigma : \tau] \alpha(\tau), \quad (1)$$

with $[\sigma : \tau] \in \{\pm 1\}$ the relative orientation. The defining identity of a chain complex is $d_{k+1} d_k = 0$.

A choice of Hermitian inner products on each C^k (here induced by the edge weights, see Remark 1) gives adjoints $d_k^\dagger : C^{k+1} \rightarrow C^k$. The *Hodge Laplacian* in degree k is

$$L_k = \underbrace{d_{k-1} d_{k-1}^\dagger}_{\text{“down”}} + \underbrace{d_k^\dagger d_k}_{\text{“up”}}, \quad L_k = L_k^\dagger \succeq 0. \quad (2)$$

Remark 1 (The magnitude convention at $k = 0$). At $k = 0$ the down-term vanishes and $L_0 = d_0^\dagger d_0$ is the graph Laplacian. With a *Hermitian* weight $A_{ij} = w_{ij} e^{i\phi_{ij}}$ on the edge $i \rightarrow j$ (so $A_{ji} = \overline{A_{ij}}$) and the *magnitude* degree $D_{ii} = \sum_j |w_{ij}|$, the project uses

$$L_0 = D - A, \quad D = \text{diag}(D_{ii}), \quad (3)$$

a Hermitian operator whose evolution $e^{-iL_0 t}$ is unitary. The phase ϕ_{ij} is a $U(1)$ connection living on the edge; its loop sums are gauge-invariant fluxes. This is exactly **cobordism::HodgeLaplacian** (the class is degree-parameterized: $k = 0$ from edge weights, $k \geq 1$ from the boundary matrices of **ChainComplex**).

The central fact is the discrete Hodge theorem.

Definition 2 (Harmonic cochains). $\alpha \in C^k$ is *harmonic* iff $L_k \alpha = 0$; equivalently $d_k \alpha = 0$ and $d_{k-1}^\dagger \alpha = 0$ (closed and co-closed).

Theorem 1 (Discrete Hodge decomposition). $C^k = \text{im } d_{k-1} \oplus \ker L_k \oplus \text{im } d_k^\dagger$, and the harmonic space is isomorphic to cohomology,

$$\ker L_k \cong H^k(K; \mathbb{C}), \quad \boxed{\dim \ker L_k = b_k} \quad (4)$$

the k -th Betti number. Harmonic representatives are the unique minimum-norm elements of their cohomology classes.

So the *kernel* of L_k is a coordinate-free handle on topology: its dimension counts k -dimensional “holes,” and its elements are canonical (metric-orthogonalized) representatives of those classes. The eigenvectors of L_k with small nonzero eigenvalue are the “almost-harmonic” modes; the spectrum interpolates between geometry (large eigenvalues, local) and topology (the kernel, global).

4 Harmonic forms and the Hodge qubit

Take Σ a closed oriented surface of genus g . Then $b_1(\Sigma) = 2g$, so $\dim \ker L_1(\Sigma) = 2g$: there are exactly $2g$ independent harmonic 1-forms, one “dual” pair per handle.

Definition 3 (Hodge qubit). For the torus T^2 ($g = 1$) we have $\dim \ker L_1(T^2) = 2$: a two-dimensional complex vector space. We call it the *Hodge qubit*. A unit harmonic form

$$\psi = \alpha \omega_x + \beta \omega_y \in \ker L_1(T^2), \quad |\alpha|^2 + |\beta|^2 = 1, \quad (5)$$

written in the basis of the two harmonic generators ω_x, ω_y (the discrete analogues of dx, dy around the two cycles), is the qubit state $\alpha|0\rangle + \beta|1\rangle$.

The amplitudes and relative phase of a quantum state are encoded as the weights and holonomies of a harmonic 1-form on a surface. Boundary states for the cobordism are chosen as $\psi_A \in \ker L_1(\Sigma_A)$, $\psi_B \in \ker L_1(\Sigma_B)$. (The $k = 0$ analogue — a node-amplitude qubit $\ker L_0$ on a small graph — is the reduced setting used for the first realizability oracle; the surface/ $k=1$ version is the one that matches Dijkgraaf–Witten, §5–§6.)

5 Holonomy, flat connections, and Dijkgraaf–Witten

Now switch from continuous harmonic forms to discrete gauge fields. A *flat $\mathbb{Z}_2$ connection* on a complex assigns $g_e \in \mathbb{Z}_2 = \{0, 1\}$ to each edge so that the holonomy around every triangle is trivial; modulo gauge transformations these are classified by

$$H^1(\Sigma; \mathbb{Z}_2) = \text{Hom}(H_1(\Sigma), \mathbb{Z}_2) \cong (\mathbb{Z}_2)^{b_1(\Sigma)}. \quad (6)$$

A class is fixed by its *holonomy* ± 1 around each of the b_1 independent cycles, so there are 2^{b_1} of them.

Definition 4 (Dijkgraaf–Witten boundary space). A boundary surface Σ carries the Hilbert space

$$Z(\Sigma) = \mathbb{C}[H^1(\Sigma; \mathbb{Z}_2)], \quad \dim Z(\Sigma) = 2^{b_1(\Sigma)}, \quad (7)$$

spanned by the holonomy classes (one basis vector per flat connection).

For a finite group G (here $\mathbb{Z}_2$) and a group n -cocycle $\omega \in H^n(BG; U(1))$, the *Dijkgraaf–Witten state sum* of a closed n -manifold W is

$$Z(W) = \frac{1}{|G|^{|V|}} \sum_{\substack{g: \text{edges} \rightarrow G \\ \text{flat}}} \prod_{n\text{-simplices } t} \omega(g_t)^{\varepsilon_t}, \quad (8)$$

a number; $\varepsilon_t = \pm 1$ is the orientation of the top simplex t . For $n = 3$ and $G = \mathbb{Z}_2$, ω is one of two 3-cocycles — the trivial one ($\omega \equiv 1$) and the nontrivial *sign* class ($\omega = (-1)^{abc}$). For a cobordism W with boundary one holds the boundary connection fixed and sums the same product over interior flat fields; the result is a vector in $Z(\partial W)$, equivalently (for two boundary components) a linear map

$$Z(W) : Z(\Sigma_B) \longrightarrow Z(\Sigma_A), \quad \langle \psi_A | Z(W) | \psi_B \rangle = \sum_{a,b} \overline{\psi_A[a]} Z(W)_{ab} \psi_B[b]. \quad (9)$$

For the trivial cylinder $\Sigma \times [0, T]$ this map is the identity on $Z(\Sigma)$. In the codebase this is `cobordism::DijkgraafWitten: partitionFunction()` (closed), `map()` / `boundaryVector()` (with boundary), and `amplitude( $\psi_A, \psi_B$ )` for the sandwiched value above.

6 Two countings, one preparation map

We now have *two* boundary spaces attached to the same surface Σ , and they are *not* the same:

	Hodge qubit	Dijkgraaf–Witten space
object	harmonic 1-forms $\ker L_1(\Sigma)$	holonomy classes $\mathbb{C}[H^1(\Sigma; \mathbb{Z}_2)]$
dimension	b_1 (continuous, over $\mathbb{C}$)	2^{b_1} (discrete count)
torus T^2	$\mathbb{C}^2$ (one qubit)	$\mathbb{C}^4$
role	the spectral state	the state-sum basis

Remark 2 (Keep the countings distinct). b_1 versus 2^{b_1} : continuous harmonic forms give the *qubit dimension*; flat connections *index the state sum*. They coincide only trivially. This is stated as a standing convention in the experiment charter (§5.2), and the Dijkgraaf–Witten class deliberately keeps the “ b_1 qubit” separate from the “ 2^{b_1} flat-connection count.”

The two sides meet through a *preparation map*

$$\text{prep} : \ker L_1(\Sigma) \longrightarrow \mathbb{C}[H^1(\Sigma; \mathbb{Z}_2)], \quad b_1 \mapsto 2^{b_1}, \quad (10)$$

which encodes a harmonic (spectral) boundary state as an amplitude over holonomy classes; `DijkgraafWitten.amplitude()` already consumes states in the flat-connection basis that have been prepared this way. Making `prep` and its readout first-class and tested is the foundational step of the bridge work — it is the only place the spectral and topological descriptions of the *boundary* must be reconciled. (Everything “interesting” then happens in the *bulk* invariant.)

7 Choi–Jamiolkowski: operations as states (the bend)

To feed an operation into a geometry we must first turn it into a state. The vehicle is the Choi–Jamiolkowski isomorphism. Define the *vectorization* of $U = \sum_{ij} U_{ij} |i\rangle\langle j| : \mathcal{H}_B \rightarrow \mathcal{H}_A$ by stacking rows,

$$\text{vec}(U) = \sum_{ij} U_{ij} |i\rangle \otimes |j\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B, \quad \text{vec}(|a\rangle\langle b|) = |a\rangle \otimes \overline{|b\rangle}. \quad (11)$$

The operator becomes a (generally entangled) bipartite state — the *Choi state*. Two facts make it the right bend.

Theorem 2 (Hilbert–Schmidt / Choi identity). *For the rank-one transition operator $U_T = |\psi_A\rangle\langle\psi_B|$,*

$$\langle\psi_A|U|\psi_B\rangle = \text{Tr}(U_T^\dagger U) = \langle\text{vec}(U_T), \text{vec}(U)\rangle. \quad (12)$$

That is: the physical amplitude $\langle\psi_A|U|\psi_B\rangle$ equals an *inner product of Choi states*. So if a cobordism faithfully carries $\text{vec}(U)$ on its output boundary, contracting that boundary state against the test functional $\text{vec}(U_T)$ returns the amplitude — which is exactly how the spectral side reads off “ Z .” (This is `quantum::ChoiJamiolkowski: vectorize, transitionOperator, transitionAmplitude`.)

Theorem 3 (Rank = Schmidt rank = connectivity). *The Schmidt rank of $\text{vec}(U)$ equals $\text{rank}(U)$ (the number of nonzero singular values of U). A separable Choi state (rank 1) corresponds to a rank-one $U_T = |\psi_A\rangle\langle\psi_B|$ and a disconnected cup/cap geometry; higher Schmidt rank is genuine entanglement and a connected bulk.*

8 Finding topology by minimizing eigenvector residuals

We can now state the inverse problem precisely. Given a target boundary state ψ (the bent operator, $\psi = \text{vec}(U)/\|\text{vec}(U)\|$), *find a complex W whose Hodge Laplacian has ψ as an eigenvector*. The eigenvalue is a free output, so we use the eigenvalue-agnostic *residual*

$$r(W) = \|(I - \psi\psi^\dagger)L(W)\psi\|^2 = \|L\psi - (\psi^\dagger L\psi)\psi\|^2. \quad (13)$$

$r(W)$ is the squared norm of the component of $L\psi$ orthogonal to ψ ; it vanishes *iff* $L\psi = \lambda\psi$ with $\lambda = \psi^\dagger L\psi$ the Rayleigh quotient. Minimizing r over the degrees of freedom of W is thus a search for a geometry that *realizes* ψ spectrally.

Fixed-boundary fill. For realizability the boundary is *pinned*: ∂W is the synthesized $\text{geo}(\psi_A), \text{geo}(\psi_B)$ and the output surface, held fixed. The free parameters θ are the *interior* edge weights/phases and — through boundary-fixed Pachner “add” moves (coning) — the interior *topology*. This is the decisive twist: with the ends fixed the parameters do *not* outrun the constraints, so realizability becomes a genuine test rather than a foregone conclusion.

Verdict and certificate. A target is **realizable** iff r can be driven below ϵ ; it is **obstructed** iff r *floors* at a positive value — a residual lower bound under the fixed-boundary constraint. The floor is the non-existence *certificate* (the analogue of the two-vertex floor $w_{\min}^2(|c_0|^2 - |c_1|^2)^2$), obtained without exhausting triangulations.

Algorithm. DECIDEREALIZABILITY($U, d_A, d_B; \epsilon, R, c_{\max}$)

Input: operation $U : \mathcal{H}_B \rightarrow \mathcal{H}_A$; tolerance ϵ ; restart count R ; cone budget $c_{\max}$.

Output: REALIZABLE with witness W , or OBSTRUCTED with floor f .

```

 $\psi \leftarrow \text{vec}(U) / \|\text{vec}(U)\|$                                      // bend: Choi–Jamiolkowski (§7)
 $W \leftarrow \text{seed bulk}; \text{PIN}(\partial W)$                              // fixed ends; only interior edges are writable
 $f \leftarrow +\infty$ 
for  $c = 0, 1, \dots, c_{\max}$  do
  define  $r_W(\theta) = \|(I - \psi\psi^\dagger)L(W; \theta)\psi\|^2$              // residual over interior params  $\theta$ 
   $(\theta^*, r^*) \leftarrow \text{MULTIRESTARTLM}(r_W, R)$                  // Levenberg–Marquardt  $\times R$  seeds
  if  $r^* < \epsilon$  then return REALIZABLE,  $W(\theta^*)$                  // witness carries  $\psi$  as an eigenvector
   $f \leftarrow \min(f, r^*)$ 
   $W \leftarrow \text{CONEINTERIOR}(W)$                                  // boundary-fixed Pachner add: +1 interior vertex
return OBSTRUCTED, floor  $f$                                      //  $f$  bounded away from 0 certifies non-existence

```

In the codebase: `ChoiJamiolkowski` does the bend; `EigenstateSynthesis` evaluates r_W and the fixed-boundary fill; `LevenbergMarquardt` is the multi-restart solver; `RealizabilityOracle.decide()` is the loop above; the witness W is returned as a `Spacetime`, and `getBoundary()` recovers ∂W .

9 The bridge

Putting the pieces together on a single synthesized W_{AB} :

- the spectral road (§8) produces W_{AB} whose boundary harmonic carries $\text{vec}(U)$, so by Theorem 2 the *spectral* $Z_{\text{spec}}(W_{AB}) = \langle \psi_A | U | \psi_B \rangle$;

- the topological road (§5) computes $Z_{\text{DW}}(W_{AB}) = \langle \psi_A | Z(W_{AB}) | \psi_B \rangle$ from the triangulation and cocycle alone, with boundary states obtained through the preparation map of §6.

The bridge is the assertion

$$Z_{\text{DW}}(W_{AB}) \stackrel{?}{=} \langle \psi_A | U | \psi_B \rangle \stackrel{?}{=} Z_{\text{spec}}(W_{AB}). \quad (14)$$

Remark 3 (A discrete shadow). The $\mathbb{Z}_2$ Dijkgraaf–Witten functor sends cobordisms to a *discrete* family of maps (the Frobenius algebra $\mathbb{C}[\mathbb{Z}_2]$ generated by pair-of-pants, cap/cup, cylinder, twisted by the cocycle). A generic U is *not* Z_{DW} of any $\mathbb{Z}_2$ cobordism. The honest expectation, then, is that the bridge holds on the discrete *DW-representable* subset, and that the spectral oracle — which realizes a *continuum* of operators — strictly extends the topological theory. In that picture the Dijkgraaf–Witten invariant is the *quantized shadow* of the spectral Z : equal where both are defined, with the geometry seeing strictly more than the finite-group state sum.

Where this lives. HodgeLaplacian (§3–§4), DijkgraafWitten (§5–§6), ChoiJamiolkowski (§7), and RealizabilityOracle/EigenstateSynthesis/LevenbergMarquardt (§8) are in place; the preparation map of §6, the $k=1$ 3-manifold synthesis, and the agreement experiment of §9 are the three steps of the bridge milestone.